

Auditing Evidence Claims in Federal High-Impact AI Exclusions

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Abstract

Public AI registers disclose risk classifications, yet exclusion rationales may omit evidence needed for review. We examine 110 entries labeled presumed high-impact but not high-impact in the 2025 U.S. Federal Agency AI Use Case Inventory. A policy-derived instrument separates a stated predicate, an AI output and workflow role, and a mechanism limiting consequential reliance. Conservative reconciliation yields 24 minimally supported, 70 unresolved, and 16 human-review-only cases. The unresolved share falls from 63.6% of records to 50.0% of distinct rationales and 48.7% after excluding the Department of Justice. In two exact-request rounds, GPT-5.4 and GPT-5.4-mini agree across all four readings on only 45 of 70 circuit statuses and 13 component vectors. A separate 700-request test with GPT-6 Astra and GPT-5.4-mini distinguishes an explicit independent basis from weaker additions; line-wrap controls reveal instability that cannot be attributed solely to whitespace. Full-text review of seven linked record-document pairs finds candidate mechanisms in three, but no complete system-and-version support chain under a strict rule. Independent human annotations are unavailable. The result is an exploratory method for auditing missing evidence, unstable interpretation, and incomplete provenance.

Introduction

Risk classification determines which governance practices apply to an AI use. Office of Management and Budget (OMB) Memorandum M-25-21 requires minimum practices for federal high-impact AI and permits determinations that a presumptively high-impact use does not meet that definition. The inventory makes such exclusions visible. It does not necessarily make their reasoning reconstructible. A reader may see a negative determination, a general assurance about human involvement, and a privacy-assessment link without learning why the relevant threshold is not met.

The distinction matters because the policy's principal-basis test can apply with or without human decision-making (Office of Management and Budget 2025, sec. 4(a), p. 14). A human signature alone therefore does not resolve the classification question. Relevant public evidence might instead explain how an official independently reconstructs a decision from original records, how an alternative procedure avoids reliance on the AI, or how a technical limit excludes

a consequential use. Human oversight has institutional and practical weaknesses that a declaration of review cannot settle (Green 2022).

Automating this audit creates a second accountability problem. An evaluator can impose a coherent formal rule while interpreting its components inconsistently. Repeated agency language can dominate record-level percentages. A linked document can contain useful evidence while leaving the relationship to a particular AI use uncertain. Consequently, a study of public reasons must examine the stability and provenance of its own judgments.

We ask four questions. **RQ1:** What public support is visible under a declared rebuttal instrument? **RQ2:** How does the answer change with the unit of aggregation, computational reading, and treatment of identical text? **RQ3:** Do computational judges respond differently to evidence that should and should not satisfy the instrument? **RQ4:** What can complete linked documents add, and which evidentiary links remain open? Our population is the complete set of 110 disclosed rows matching one inventory category, not a census of every federal AI deployment.

The study contributes a policy-derived component model, an executable consistency audit, a prospective evidence-response test, and a record-level repair analysis. The test adds human review, generic multiple-source language, or a concrete independent basis to every distinct rationale and checks the paired response. The empirical contribution includes failure evidence. The majority-unresolved finding depends on agency composition; exact repeats and line-wrap control pairs can disagree; and useful document passages do not automatically establish system and version applicability. These results support an exploratory audit method and concrete disclosure improvements. They do not establish legal error rates or validate an autonomous policy adjudicator.

Related Work

Algorithmic transparency can expose information without making its consequences understandable (Ananny and

Crawford 2018). Research on algorithm registers examines institutional usefulness and disclosure completeness (Nieuwenhuizen 2024; Peljto, Heilmann, and Cerrato 2026). In the closest federal study, Lurie et al. (2026) compare AI inventories with System of Records Notices and Information Collection Requests, while also examining privacy assessments among the documentary materials. Their work motivates treating document identity as a substantive evidentiary question. Our narrower unit is the reason supplied for an exclusion from a presumed risk category, and our central evaluation concerns the stability of coding that reason.

Internal auditing frameworks organize accountability throughout system development (Raji et al. 2020). Assurance cases connect claims, evidence, warrants, assumptions, and challenges; AI ethics assurance patterns already formalize much of this structure (Porter et al. 2024). Work on assurance defeaters further explains how counterconsiderations can undercut otherwise plausible arguments (Gohar et al. 2025). We adapt these ideas to a public administrative record. The contribution is an application and measurement study, not a new general theory of argumentation or assurance.

Our use of language models also requires a validation boundary. Tests for replacing human annotators require evidence connecting model judgments to the relevant human judgments (Calderon, Reichart, and Dror 2025). Automated assurance-case review has likewise retained a role for human reviewers even when predicate-based LLM judges perform well (Yu et al. 2026). We have no completed independent human labels. Computational agreement, synthetic conformance, and agreement with an earlier computational panel are therefore reported separately; none is called criterion accuracy on the inventory.

Recent evaluations give a second reason to test judges directly. Rubric text can predict judge behavior even without the candidate response, and judges do not always update under counterfactual changes (Bagaria et al. 2026). Repeated-call analyses also distinguish prompt effects from execution noise and warn that a few calls do not identify a judge's long-run response distribution (Feng et al. 2026). Our exact repeats and paired evidence additions are direct diagnostics for these problems. They do not estimate a general response law.

A targeted review through September 8, 2026 identified adjacent work on registers, cross-regime linking, assurance-case judging, and counterfactual evaluation. We found no study of this population that combines rationale deduplication, composition sensitivity, exact and semantic invariance tests, controlled evidence additions, and passage-level provenance repair. This is a bounded novelty claim. The reusable contribution is the combined audit architecture and its documented failure modes; the corpus alone does not establish methodological novelty.

Public Rebuttal Instrument

Components and Status Logic

For each rationale, P denotes a recognizable policy predicate being denied, O an identified AI output together with its workflow role, and W a qualifying mechanism limiting consequential reliance. The locked codebook distinguishes an independent non-AI evidentiary basis W^{ind} , a usable non-AI alternative W^{opt} , and a narrow technical or procedural role W^{nar} . H records whether human approval, discretion, or review carries the exclusion argument. The six independently coded components are:

$$v_i = (P_i, O_i, H_i, W_i^{\text{ind}}, W_i^{\text{opt}}, W_i^{\text{nar}})$$

Each component takes YES, NO, or UNCLEAR and is accompanied by a load-bearing excerpt and an ambiguity note. A component is Boolean true only when coded YES; negation means absence of a YES finding, not proof that the underlying fact is false. The aggregate mechanism is:

$$W_i = W_i^{\text{ind}} \vee W_i^{\text{opt}} \vee W_i^{\text{nar}}$$

$$S_i = 1[P_i \wedge O_i \wedge W_i]$$

$$D_i = 1[H_i \wedge \neg W_i]$$

All remaining cases receive U. S means visible minimum support. D marks a human-review-only objection: review carries the argument, but no qualifying mechanism is established. U collects incomplete or conflicting readings. Neither D nor U establishes unlawful classification; an S rationale may still describe a mechanism inaccurately or omit upstream effects.

The codebook also requires U when two reasonable readings yield different statuses. Let C_i be the circuit status and let A_i mark an adjudicated ambiguity. The intended adjudicated semantic status T_i^* can be written as:

$$T_i^* = \begin{cases} U, & \text{if } A_i = 1 \\ C_i, & \text{if } A_i = 0 \end{cases}$$

The archived responses contain ambiguity notes, but no consistently adjudicated Boolean A_i , so this expression does not recode them. We denote an observed model-reported status by R_i and retain it separately from circuit status C_i , intended adjudicated status T_i^* , the archived panel vote, and the duplicate-reconciled reference. $H=\text{YES}$ with no $W=\text{YES}$ gives D mechanically even when a possible W is UNCLEAR; a model may report U under the ambiguity rule.

A mechanism connects a named constraint to the output's influence. General claims of assistance, validation, or multiple inputs do not suffice. A manual alternative must avoid dependence; its existence does not prove its use. A narrow role must explain the relevant bound. Development stage is separate because a pilot limitation can cease after deployment. Negation and clause attachment are assessed within the statement without combining disconnected assurances.

A worked example is TREAS-94: “AI Contract Document Toolbox drafts procurement narratives, supports vendor evaluations, and aids source selection. It offers recommendations but does not approve contracts. Final decisions are made by agency officials using multiple information sources.” The reconciled labels are P=YES, O=YES, H=YES, $W_{\text{ind}}=\text{UNCLEAR}$, $W_{\text{opt}}=\text{NO}$, and $W_{\text{nar}}=\text{NO}$. The last phrase identifies no independent source or constraint on reliance. Thus $W=0$, $S=0$, and $D=1$; the circuit status is D.

Estimands and Uncertainty

Our primary descriptive quantity is the fraction of disclosed records assigned each status by a specified instrument. Conditional on the archived data and deterministic aggregation rule, this fraction is exactly calculable. A template-weighted fraction instead describes distinct rationales; an equal-agency average describes the mean of agency-specific fractions. These quantities answer different questions. We report them together without treating one as a statistically corrected estimate of another.

We do not infer a latent proportion of legally correct exclusions or of exclusions supportable by undisclosed evidence. In particular, the interval from the S fraction to one minus the D fraction would require assumptions that S judgments are sound and that D cases cannot become supported. The second assumption conflicts with evidence repair: a D case can acquire an independently documented mechanism. Without a validated measurement link, that interval is not justified. We therefore report coded proportions and specification changes, with no purported logical bound on underlying correctness and no sampling confidence interval for this finite disclosed population.

Data and Computational Protocol

Corpus and Archived Readings

We use the 3,611-row CSV from OMB's 2025 Federal Agency AI Use Case Inventory (Office of Management and Budget 2026). The archived filename is:

2025_individually_reported_AI_use_cases.csv

The repository commit is:

06c7ebee5b376524211042bd9673b2a7fef3e3

The supplement records the full SHA-256. Exact filtering on “Presumed High-Impact, but Not High-impact” in `is_high_impact` selects 110 records from nine agencies. Thirteen blank public identifiers receive deterministic hashes derived from agency and use-case name. Original identifiers, rationales, source hashes, and row mappings are retained.

The existing codebook was hashed before the archived five-pass coding runs; its SHA-256 prefix is 0a1dffe440a2. Those passes emphasize literal entailment, policy mapping,

linguistic scope, adversarial false positives, and assurance-case reconstruction. Prior labels and other passes' answers were omitted from the prompts. The archived prompts nevertheless contain record context, so label isolation is not equivalent to rationale-only input.

A four-of-five status vote gives the archived reference partition, with U when no status reaches the threshold. Components use majority voting, with UNCLEAR on ties. The status vote is not recomputed from majority components: composing component majorities can produce an argument that fewer than four complete readings endorse. All individual-pass statuses are checked against their own component circuit.

The provider model identifiers for these 550 archived readings were not recorded. Their saved prompts and responses permit inspection and deterministic reaggregation, but exact judgment regeneration is limited. We retain this defect explicitly and add new identifiable runs. The 110-row human-validation sheet has no completed coder or label fields. It is an unused annotation instrument, not a human study.

Identical Text and Composition Sensitivity

Post-review, we group rationales by exact text after whitespace normalization, preserving case and punctuation. There are 70 groups, including one blank group, and 69 distinct nonempty rationales. This operation does not merge paraphrases or infer that systems themselves are equivalent.

For a rationale-only construct, identical text should receive the same label. We apply a conservative consistency rule to the archived results: a group retains its status only if all occurrences have that status; otherwise every occurrence receives U. Conflicting component labels become UNCLEAR. Original values remain in separate columns. This deterministic screening rule exposes disagreement; it is not expert adjudication or a claim that U is the semantically correct answer.

We calculate record and template partitions, agency-specific counts, leave-one-agency-out results, equal-agency averages, and deployment-stage partitions. We also repeat the text-consistency screen at three-, four-, and five-vote thresholds. These are post-review sensitivity analyses, not a pre-registered search for a preferred percentage.

Identifiable Rationale-Only Replication

We run a computational replication with GPT-5.4 and GPT-5.4-mini on September 6, 2026 UTC. The resolved identifiers are recorded on separate lines below.

gpt-5.4-2026-03-05

gpt-5.4-mini-2026-03-17

Each request supplies the unchanged codebook, one rationale, and an opaque identifier. Agency, title, previous la-

bels, and other records are withheld. Tools are disabled; reasoning effort is medium and the output ceiling is 12,288 tokens. Prompts, responses, stop conditions, usage, and hashes are saved. The scorer derives status mechanically from components. After seeing the outputs, we add a reported-status sensitivity analysis to retain possible ambiguity overrides.

Each model receives 70 inventory templates, 50 constructed cases, eight external-source challenges, and ten unchanged-text repeats: 138 requests per model and 276 total. Repeat selection was fixed before the completed run: the known discordant template plus nine templates selected by a deterministic hash order. The repeat uses a fresh opaque identifier. It measures unchanged-text stability in a small diagnostic set, not a population error rate or a fully identical-input test.

After examining that run, we save a second protocol and analysis plan, then repeat all 70 inventory requests once per model on September 6, adding 140 responses. Every prompt is byte-identical to its first-round counterpart, including the opaque identifier; the model and supplied settings are unchanged. The plan was fixed before this repeat run, but after related results had been examined; it was not externally pre-registered. We report status and six-component vector agreement, and unanimity across two models by two rounds. Provider execution state remains uncontrolled. Two observations per template do not estimate long-run reliability.

To localize status disagreements, we also enumerate all inclusion-minimal sets of coded components whose replacement with the other reading's values changes the circuit to that reading's status. Both directions are inspected. These substitutions diagnose the scoring rule; they are neither edits to the rationale nor causal experiments on model behavior. No inventory label is recoded from these diagnostics.

An exhausted 36-response pilot is retained and excluded; the whole run was restarted under the reported settings. Both completed models share a provider and family, so agreement cannot establish independent validation. Differences from the archived panel also combine context, aggregation, and model conditions.

Constructed and External Challenges

The existing synthetic suite has 50 codebook-derived rationales, including three explicitly contested cases excluded from conformance scoring. Expected component labels were assigned during construction; they are not independent expert judgments. An existing lexical comparator and the two new semantic models are evaluated against these expectations. This checks implementation behavior on known constructions, not accuracy on real records.

To add a source outside the inventory, we use the U.S. Office of Personnel Management's (OPM) August 27, 2026 hiring guidance (Office of Personnel Management 2026, PDF pp. 2-4). It distinguishes independent reasoning from

human approval alone and gives examples involving hiring-content preparation, independent applicant review, bounded pre-offer quality checks, and aggregate analysis. Before the completed replication, we created four paraphrases of these workflows and four paired controls that remove the limiting mechanism while retaining a human-approval assertion. The codebook-derived expectations are S for the four mechanism cases and D for the four controls. The prompts contain neither these labels nor source metadata.

These are source-anchored challenges, not official OPM classifications or a representative benchmark. The controls are constructed counterfactuals. Because the guidance post-dates the inventory, it challenges the rubric without redefining the 2025 standard.

Prospective Evidence Response Test

After reviewing the replication, we save five versions of each of the 70 distinct rationales and a paired scoring plan before making the 700 test requests. The original is joined by a line-wrap control and three fixed additions. The first says that a human reviews the AI output and makes the final decision. The second says that the decision maker considers multiple sources. Under the codebook, neither statement alone establishes an independent basis. The third says that the responsible official checks relevant facts against original non-AI records and permits action only on that evidentiary basis. It is designed to satisfy W^{nd} while leaving any missing P or O premise open. An addition does not automatically override the original rationale: the judge codes the combined text under the ordinary contradiction and ambiguity rules.

Each version receives a new opaque identifier; the prompt withholds the condition and paired cases. GPT-6 Astra and GPT-5.4-mini each receive all 350 inputs as separate tool-free requests on September 8, 2026 UTC. Their resolved identifiers are:

gpt-6-astra

gpt-5.4-mini-2026-03-17

We use low reasoning effort, a 4,096-token ceiling, and a strict JSON schema. The saved plan specifies invariance under line wrapping; no new qualifying W or non-S-to-S transition from human review; no new W^{nd} or non-S-to-S transition from generic multiple-source language; and $W^{nd}=YES$ after the explicit independent-basis addition. Because every pair also changes its identifier and uses separate execution, observed differences do not isolate line wrapping as their cause. These constructed-text tests do not add facts to the federal records or create human ground truth.

Results

The Aggregate Depends on Its Unit

Conservative text reconciliation changes one Department of

Homeland Security (DHS) record, DHS-2543, from S to U. The resulting partition is 24 S, 70 U, and 16 D, compared with the archived 25, 69, and 16. The reconciled unresolved share is 63.6% of records. Table 1 shows why this cannot be generalized as a stable majority of distinct public arguments.

Population	S	U	D	U share
All records, n=110	24	70	16	63.6%
Without Justice Dept., n=78	24	38	16	48.7%
Templates, n=70	22	35	13	50.0%
Nonempty, n=69	22	34	13	49.3%

Table 1: Reconciled four-vote partition by aggregation unit. S is minimally supported, U unresolved, and D a human-review-only objection. The blank rationale is retained in the census and separated in the nonempty-template row.

All 32 Department of Justice (DOJ) records repeat one rationale and receive U. Removing this agency changes the substantive majority claim, while leaving the census percentage arithmetically correct. Across all leave-one-agency-out analyses, the U share ranges from 48.7% to 78.7%; excluding DHS gives the upper endpoint. The equal-agency mean is 67.7%, but small agencies receive as much weight as large ones. It is a different estimand, not evidence that agency composition is irrelevant.

Stage checks also have different denominators. Among 61 deployed records, 41 are U (67.2%). The pilot, pre-deployment, and retired groups contain 5/10, 24/36, and 0/3 U records, respectively. These are descriptive strata, not independent samples. Text-consistent three-, four-, and five-vote partitions yield S/U/D counts of 31/59/20, 24/70/16, and 21/77/12. Every threshold leaves a record-level majority unresolved, but threshold stability does not resolve template or agency dependence.

Identical Rationales Reveal Disagreement

Three repeated-text groups contain component or status conflicts. DHS-123 and DHS-49 are both D but disagree on O. DHS-125, DHS-2427, and DHS-48 are all S but disagree on the narrow-role component; they retain a shared independent-basis mechanism. DHS-2543 and DHS-372 have exactly identical raw rationales and identical majority component labels, yet the former is unanimously S while the latter receives only three S votes and consequently U under the four-vote rule.

The disputed rationale describes collated employee information that supports staffing priorities while excluding security and employment decisions. The interpretive question is whether this establishes a bounded administrative role or leaves a consequential staffing pathway open. GPT-5.4 assigns U and then S on exact-request repetition;

GPT-5.4-mini assigns S and then U. Both models also reversed in the preliminary ten-case test. The exact repeats remove the changed-identifier explanation for this instability. Enforcing shared U in the reference table removes an arbitrary difference between occurrences, but it does not resolve that substantive ambiguity.

The archived five readings have mean pairwise status agreement .907 and Fleiss $\kappa=.845$. Those figures describe correlated computational consistency. The duplicate example demonstrates that they cannot substitute for a direct invariance check or independent expert validation.

Fresh Readings Are Not a Validation of the Panel

All 276 initial replication responses and 140 additional exact-repeat responses contain valid six-component labels. Table 2 reports the inventory partitions, with one template judgment mapped to every occurrence. This enforces within-run text consistency by construction; repeat requests test whether fresh judgments remain consistent.

Reading	S/U/D records	S/U/D templates
Reconciled panel	24 / 70 / 16	22 / 35 / 13
GPT-5.4 first	34 / 59 / 17	32 / 24 / 14
Mini first	32 / 61 / 17	29 / 27 / 14
GPT-5.4 repeat	35 / 57 / 18	32 / 23 / 15
Mini repeat	38 / 55 / 17	35 / 20 / 15

Table 2: Inventory partitions for 110 records and 70 templates. The archived panel row uses the four-of-five status vote; identifiable model rows use component-derived circuit scoring. Mini denotes GPT-5.4-mini. The two new rounds use identical prompts and API settings.

In the first round, the models agree on 52/70 template statuses (74.3%) and only 22/70 complete component vectors (31.4%). Thus 30 shared statuses conceal a disagreement about at least one premise. They agree with the reconciled panel on 55/70 and 53/70 statuses, respectively; all three readings agree on 46 templates. Neither comparator is ground truth. The first-round models leave 59/110 and 61/110 records unresolved, but only 27/78 and 29/78 after DOJ removal. Their template-level U shares are 34.3% and 38.6%.

Table 3 separates conformance on constructed expectations from stability under exact repetition.

Diagnostic	GPT-5.4	GPT-5.4-mini
Constructed statuses	42/47	42/47
O match	34/47	33/47
External challenges	8/8	8/8
Repeat circuit status	62/70	57/70
Repeat reported status	59/70	57/70
Exact repeat full vector	40/70	34/70

Table 3: Conformance and exact-request stability are different outcomes. Constructed and external expectations are not independent human labels. Repeat denominators cover all 70 templates.

The census repeat is more stable than the selected ten-case diagnostic, which gave circuit-status agreement of 6/10 and 7/10. Nevertheless, 8/70 and 13/70 exact requests change circuit status. GPT-5.4 retains exactly 32 S templates in both rounds while changing eight individual labels, demonstrating why aggregate stability is insufficient. Only 45/70 templates have one circuit status across all four readings: 21 S, 16 U, and eight D. Seven of the 52 initial between-model agreements fail this repeated agreement check. Complete component vectors agree across all four readings on only 13/70 templates. These are observed stability subsets, not validated labels or a new consensus estimator.

The narrow-role component is a major source of interpretive sensitivity. It differs between models on 26/70 first-round templates, versus 17 for independent evidence, nine for human review, eight for output/workflow, seven for alternatives, and four for predicate. Among the 18 first-round status disagreements, replacing only the narrow-role label is sufficient to reproduce the opposite status in at least one direction for 13. The corresponding counts are 7/8 GPT-5.4 repeat disagreements and 5/13 mini repeat disagreements. A stated limitation such as assistance or a nonfinal role therefore warrants particular scrutiny, but these circuit diagnostics do not decide which reading is right.

Both models satisfy all eight source-anchored challenges, but each matches only 42/47 synthetic status expectations. Their O component conformance is lower still. The lexical comparator matches 43/47 statuses while matching O on only 30/47. These figures do not show that lexical coding is more accurate on real data; they show that the suite's status score can conceal substantial component disagreement.

The ambiguity override materially affects D/U interpretation. GPT-5.4 reports U where the circuit gives D in seven initial responses, including four inventory templates, and five exact repeats. All twelve contain ambiguity notes; these differences are compatible with the codebook's override and should not be described simply as model errors. Using reported statuses gives GPT-5.4 template counts of 32 S, 28 U, and ten D in both rounds, despite eleven template-level

status changes. Its mapped record counts become 34/64/12 initially and 35/63/12 on repetition. Mini's reported and circuit statuses coincide. Four-reading status unanimity is 43/70 using reported labels, so the instability conclusion survives the scoring convention.

Exact-substring checks flag 12 and 31 nonliteral excerpt fields in the initial runs, and three and 25 in the exact repeats, respectively. These may include formatting or paraphrase errors and are retained for inspection, not treated as semantic ground truth. Syntactically complete responses alone cannot establish correct or faithfully quoted judgments.

Evidence Additions Separate Rubric Response from Accuracy

All 700 prospective responses satisfy the output schema. The saved plan tests human review against any new W, generic sources against new W^{ind} , and both additions for non-S-to-S transitions. Table 4 marks these endpoints [P] and additional post-review diagnostics [A]. Base-no-W and base- W^{ind} denominators lack a YES on the corresponding mechanism; non-S includes every original status other than S. Eligible-S further requires original P=YES, O=YES, and no W=YES.

Diagnostic	Astra	GPT-5.4-mini
Line-wrap vector; status [P]	60/70; 69/70	38/70; 61/70
Human new W; non-S to S [P]	1/53; 1/53	4/51; 4/52
Generic new W^{ind} ; non-S to S [P]	1/63; 1/53	2/62; 6/52
Generic new W [A]	1/53	6/51
Human; generic eligible S [A]	1/31; 1/31	4/33; 6/33
Explicit W^{ind} =YES [P]	70/70	70/70
Explicit eligible S [A]	31/31	31/33

Table 4: Paired evidence-response results. [P] marks a saved prospective endpoint and [A] an additional post-review diagnostic. Semicolon-separated values follow the label order. Departures diagnose judge behavior, not legal error or a uniquely identified wording effect.

Both judges assign W^{ind} =YES to the explicit version in 70/70 cases. Among eligible originals, it yields S for 31/31 Astra cases and 31/33 mini cases. In the two mini exceptions, O changes from YES in the original reading to NO or UNCLEAR in the paired reading, although the added sentence does not change the original output description; the circuit therefore remains U. The transformation can affect more than its target component.

The independent-evidence wording is behaviorally distinguishable from generic oversight language for both models, without validating its application to the inventory. Judge choice remains consequential: the line-wrap pairs preserve 69/70 Astra circuit statuses and 61/70 mini statuses, while exact component vectors are less stable. A stronger causal test would hold identifiers fixed, repeat both unmodified and wrapped inputs, and compare the paired difference with unchanged-input variation. These failures show why an added model vote cannot replace expert annotation.

Repairs Separate Disclosure from Interpretation

Using reconciled component labels, we compute missing roles in the minimum-support circuit for each of the 70 U records. Twelve need one role, 43 need two, and nine need all three. The remaining six have no component-level deficit but remain U because complete readings or duplicate occurrences conflict. Their route is interpretation review, not additional boilerplate.

Across U records, 63 lack a qualifying mechanism, 49 lack an output/workflow relation, and 13 lack a recognized predicate. Counts overlap. A repair set describes what would have to be supplied and verified; it does not assert that a missing fact exists. D records can also be repaired if credible evidence answers the human-review-only objection. This is why D is unsuitable as a permanent exclusion from an upper bound on eventual support.

Full-Document Follow-Up

Coverage and Evidence Rule

The original link analysis contains 22 record-link pairs but only 18 distinct URLs. Nine linked records are U in the archived partition; the other linked records are 12 S and one D. The prior computational passes saw selected excerpts capped at 22,000 characters. Their zero-completion result therefore described an excerpt-based procedure, not exhaustive full-document recovery.

We follow up all nine original U cases. Six record links already point to PDFs; a seventh PDF is recovered behind DHS-183's landing page. Because DHS-81 and DHS-95 share a PDF, this yields seven record-document pairs and six distinct PDFs. The two remaining links are collections, for DHS-401 and a Department of Health and Human Services, Administration for Children and Families (HHS/ACF) case. Collection inspection and bounded searches using the case name and available system identifier do not identify a matching document. We preserve the search ledger and do not substitute a different system's assessment.

For each pair, the analysis uses the two September 6 mod-

els identified above, GPT-5.4 and GPT-5.4-mini. Each receives inventory context and complete page-indexed text, including appendices. Inputs range from 36,762 to 172,850 characters, without keyword windows or a cap. Five conditions are coded: same AI use, 2025 applicability, same predicate, a qualifying mechanism, and an exact passage anchor. Every positive condition must appear in both readings; quotations are checked against the cited PDF page after whitespace normalization.

The time requirement is deliberately strict: an older document alone neither proves nor disproves that its workflow applies to the 2025 entry. We therefore also report a sensitivity analysis omitting the temporal requirement. The audit inspects full extracted text, not every visual feature or an exhaustive search of all documents that might exist online.

Mechanism Evidence and Provenance Gaps

All 14 document responses contain the five semantic labels. Four omit the redundant retrieval field; the scorer uses observed retrieval and preserves this schema omission. Both models identify candidate mechanisms in three pairs: DHS-183, DHS-24, and DHS-310. For two pairs, DHS-183 and DHS-310, both mechanism and passage-anchor quotations pass the literal page check. One DHS-24 passage-anchor quotation fails that check, despite both models assigning YES to the field. Table 5 summarizes the resulting evidence chain and distinguishes model assertions from verifiable quotations.

Full-text outcome	Pairs
Complete PDF text examined	7
Both models mark mechanism YES	3
Verified mechanism and anchor	2
Complete strict support chain	0
Chain with time condition omitted	1

Table 5: Seven pairs represent six distinct PDFs. Two additional U cases lack an identified matching document after bounded follow-up.

The social-media assessment linked from DHS-183 describes corroboration using news reports, agency holdings, and other governmental sources (U.S. Customs and Border Protection (CBP) 2019, PDF p. 13). This is relevant mechanism evidence, but the assessment does not establish that it governs the inventoried Dataminr use. The ACE assessment linked from DHS-24 describes conditional manual review when information is missing, mismatched, or inaccurate, and states that facial matches are not the sole basis for admissibility (U.S. Customs and Border Protection 2023a,

PDF pp. 11 and 20). Neither statement alone proves that AI is not a principal basis, and the mapping to the inventoried Entity Resolution use remains uncertain.

DHS-310 is the strongest candidate. Its broker-management assessment describes the use of AI for remote exam proctoring, review of recorded footage, and corroboration before score invalidation (U.S. Customs and Border Protection 2023b, PDF pp. 10-11). Both models recognize the specific use and a mechanism, but neither establishes the document version's applicability in 2025. It is the one pair that completes the checked chain when the temporal condition is omitted. Upstream effects of an AI alert, such as selecting a person for scrutiny, still merit evaluation; a limitation on the final action does not automatically settle them.

Thus zero strict recovery is a finding about an explicit provenance rule over seven inspected pairs. It does not mean the linked materials contain no relevant mechanisms, that all nine U cases lack support somewhere, or that any exclusion is legally wrong. The relaxation result shows how strongly the conclusion depends on version evidence.

Implications and Limitations

Future registers should identify the denied threshold, AI output and workflow step, limiting mechanism, and supporting document version and passage. A rationale based on human review should also state the reviewer's evidence and authority and how the workflow avoids consequential dependence. Temporary rationales need a reconsideration trigger.

Auditors should separate records from repeated arguments, enforce identical-text consistency, and examine exact repetitions beside aggregate percentages. Agreement matters at the premise level as well as the final label. Evidence additions test rubric response; line-wrap controls reveal fragility; component substitutions locate distinctions needing adjudication. Narrow-role interpretation, ambiguity overrides, and quotation checks deserve particular attention.

No policy or assurance expert has validated the real-record labels. Identifiable runs and failure tests do not repair that gap. The next step is blinded expert coding stratified by agency, duplicate group, and model disagreement, followed by adjudication. Until then, the labels support exploratory diagnosis and review prioritization rather than automated compliance decisions.

The mechanism families are an operational policy interpretation, and judgments may confuse nonexclusive with nonprincipal reliance. Repeated text can hide use-specific differences, and the corpus excludes unreported uses. All identifiable runs use one provider; Astra is an alias rather than a dated snapshot. Two exact observations do not establish future reliability. Constructed cases and source-anchored paraphrases may encode the rubric, text extraction can miss visual content, and bounded searches cannot prove

absence. The exact-repeat and evidence-response plans were saved before their respective requests but after related results; neither was externally preregistered. Reconciliation, component substitution, the [A] diagnostics, and temporal ablation are post-review.

Conclusion

The instrument assigns 24 records to minimum support, 70 to unresolved, and 16 to a human-review-only objection. Agency removal and rationale deduplication change that headline result. Exact repeats and line-wrap controls reveal unstable computational readings; controlled additions distinguish explicit independent evidence from generic oversight without validating either judge. Linked documents recover candidate mechanisms but leave identity or version provenance open. The audit therefore identifies missing premises and cases for expert review rather than legal error rates.

Data and Code Availability

The source snapshot, codebook, analysis scripts, prompts, saved model outputs, row mappings, saved protocols, and supporting audit materials are available from the authors upon reasonable request.

References

- Ananny, M.; and Crawford, K. 2018. Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society* 20(3): 973-989. doi:10.1177/1461444816676645.
- Bagaria, A.; Sundaram, S. S.; Krishnan, G. S.; and Ravindran, B. 2026. Judging LLM-as-a-Judge: Concerning Rubric Artifacts in LLM-based Automated Text Generation Evaluation. arXiv:2609.02942.
- Calderon, N.; Reichart, R.; and Dror, R. 2025. The Alternative Annotator Test for LLM-as-a-Judge: How to Statistically Justify Replacing Human Annotators with LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers*, 16051-16081. Vienna: Association for Computational Linguistics. doi:10.18653/v1/2025.acl-long.782.
- Feng, P.; Meng, C.; Qu, D.; Zhang, Z.; Liu, H.; and Wu, J. 2026. Second-Order Response Laws for LLM Judges: Debaised Estimation of Prompt Instability. arXiv:2608.16253.
- Gohar, U.; Hunter, M. C.; Cohen, M. B.; and Lutz, R. R. 2025. A taxonomy of real-world defeaters in safety assurance cases. In *2025 IEEE/ACM Workshop on Multi-disciplinary, Open, and Relevant Requirements Engineering (MO2RE)*, 3-9. doi:10.1109/MO2RE66661.2025.00007.
- Green, B. 2022. The flaws of policies requiring human oversight of government algorithms. *Computer Law & Security Review* 45: 105681. doi:10.1016/j.clsr.2022.105681.
- Lurie, E.; Fauser, E.; He, Q.; Metaxa, D.; and Friedler, S. A. 2026. Triangulating across U.S. federal AI transparency regimes.

arXiv:2607.29540.

Nieuwenhuizen, E. 2024. Algorithm registers: A box-ticking exercise or meaningful tool for transparency? *Information Polity* 29(4): 415-433. doi:10.1177/15701255241297107.

Office of Management and Budget. 2025. M-25-21: Accelerating federal use of AI through innovation, governance, and public trust. Washington, DC: Executive Office of the President. <https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-21-Accelerating-Federal-Use-of-AI-through-Innovation-Governance-and-Public-Trust.pdf>.

Office of Management and Budget. 2026. 2025 Federal Agency AI Use Case Inventory. <https://github.com/ombegov/2025-Federal-Agency-AI-Use-Case-Inventory>. Archived at commit 06c7ebee5b376524211042bd9673b2a7fed3e3, May 14, 2026.

Office of Personnel Management. 2026. Use of Artificial Intelligence in the Federal Hiring Process. Memorandum, August 27. <https://www.opm.gov/chcoc/latest-memos/use-of-artificial-intelligence-in-the-federal-hiring-process.pdf>. Accessed September 6, 2026.

Peljto, I.; Heilmann, X.; and Cerrato, M. 2026. Are algorithm registers transparent? Perspectives from Germany. arXiv:2606.02347.

Porter, Z.; Habli, I.; McDermid, J.; and Kaas, M. 2024. A principles-based ethics assurance argument pattern for AI and autonomous systems. *AI and Ethics* 4(2): 593-616. doi:10.1007/s43681-023-00297-2.

Raji, I. D.; Smart, A.; White, R. N.; Mitchell, M.; Gebru, T.;

Hutchinson, B.; Smith-Loud, J.; Theron, D.; and Barnes, P. 2020. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 33-44. doi:10.1145/3351095.3372873.

U.S. Customs and Border Protection. 2019. Publicly Available Social Media Monitoring and Situational Awareness Initiative. Privacy Impact Assessment DHS/CBP/PIA-058, March 25. <https://www.dhs.gov/sites/default/files/publications/privacy-pia-cbp58-socialmedia-march2019.pdf>.

U.S. Customs and Border Protection. 2023a. Automated Commercial Environment (ACE) Modernization. Privacy Impact Assessment Update DHS/CBP/PIA-003(c), September 26. https://www.dhs.gov/sites/default/files/2023-09/23_0926_privacy-p%E2%81%AEia-cbp003c-acemodernizations.pdf.

U.S. Customs and Border Protection. 2023b. CBP Broker Management Program. Privacy Impact Assessment DHS/CBP/PIA-077, March 31. <https://www.dhs.gov/sites/default/files/2023-04/privacy-pia-cbp077-bmp-march2023.pdf.pdf>.

Yu, G.; Sivakumar, M.; Belle, A. B.; Ghari, S.; Wang, S.; and Lethbridge, T. C. 2026. LLMs as judges: Toward the LLM-assisted review of GSN-compliant assurance cases. *Journal of Systems and Software* 242: 113036. doi:10.1016/j.jss.2026.113036.